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1. Introduction

The experience of creating a covert emergency-assistance app called SilentHelp for my Bachelor's degree in Software Engineering final year project is reviewed in this paper. I discuss my actions, my responses, what I achieved, what I Failed at and what I intend to change by utilizing Gbb's reflective cycle. Gibb's framework is based on description, emotions, evaluation, analysis, conclusions and action plan.

Schon's reflection-in-action and reflection-on-action have also been applied to expose how and when I changed my decisions in the process of development and testing of the system.

The main thing I learned from this experience is that the evaluation of an application devoted to safety cannot be conducted on the basis of its interface only.

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2. Context and description

SilentHelp was developed as a stealthy emergency application with a calculator-like interface and hidden SOS functionality. The designed program covers trusted contacts, SOS triggering, geolocation and location services, messaging, continuous operation, and support for voice/sensor triggers. I opted for Flutter and Dart to develop the program instead of exploring Firebase authentication, Firestore features, and other backend services. Flutter provides an opportunity to collaborate with native platforms while implementing cross-platform features. This is important as many functions of SilentHelp require compliance with the shotoff automation, Android lifecycle and permissions, and service rules instead of relying on Flutter alone.

2.1 Week 1 – Project direction and repository setup

In the course of the project, I set an application development encironment, documented the process, launched the whole project on GitHub and invited my supervisor as a collaborator. I also wrote the introduction, acknowledgment, and abstract. The main problem I faced was the scope of the program.

2.2 Week 2 – Project elaboration and UI/UX design

I outlined the objectives, the project scope, the user flow, and the key screens as per my superior's requirements and ambition to design the project and the UI/UX. I looked into the hidden emergency evacuation route and the hidden calculator. I found out that usability and privacy are contradictory design requirements. The functionality of the feature was to be hidden but kept working.

2.3 Week 3 – Requirements and architecture

After clarifying the functional and non-functional needs, I divided the solution into services that are responsible for authentication, authorization, geo-locating, sensing, and relating the voice and SOS. I began splitting Android-specific tasks from Flutter-specific tasks. As a result, I could not regard the UI as the center of the system.

2.4 Week 4 – Flutter foundation

I have created components that are of importance in service-oriented architecture and implemented a Flutter framework. This led to easier debugging and ideal separation of concerns. However, I encountered several issues due to the need to run Flutter commands directly form the main repository rather than from the directory of the Flutter project.

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Therefore, I learned that one should first solve issues in the environment and only then make changes to the application code.

2.5 Week 5 – Calculator disguise and SOS interaction

I worked on the issue of deploying SOS functionality and developed the idea of ensuring that the app was as easy to use as a calculator was. The idea was to come up with a solution that would prevent the accidental deployment of SOS. The solution I came up with was to enable the sending of SOS as a way of changing the app's state.

2.6 Week 6 – Permissions, sensors and location

In my work, I dealt with location services, sensor logic, and permissions. I learned that sensitive permissions should not be asked for arbitrarily but always tied to a specific user goal. I also took into account the edge cases when the location, connectivity, or other prerequisites were nonexistent.

2.7 Week 7 – Background execution

The hardest requirement for me was working in the background. I researched the limitations of Android and worked on the SOS functionality in the foreground. Android has very strict requirements for using foreground services that are used for crucial resources and generally prevents execution in the background.

2.8 Week 8 – Voice trigger

I developed a speech service and explored voice-initiated activation technology. Upon initial consideration of LLM and AI/ML approaches, I realized that speech recognition and LLM tackle distinct challenges. In other words, an emergency command option is better utilised in a reliable recognition endeavour rather than unnecessarily using generative AI.

2.9 Week – Firebase and security

I analysed Firestore and Firebase Authentication security. As the development process indicates, starting your app form the right Flutter project root is very important due to configuration errors. What is more essential is the fact that Firebase integrates authentication and security rules for access control as authentication on its own does not guarantee the security of information.

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2.10 Week 10 – Offline resilience

Moreover, I analysed user behaviour in offline mode, which allowed me to differentiate between mandatory local emergency actions and optional cloud synchronisation. It dawned on me that Firebase should not be deemed as a service whose functioning is connected with every emergency component of the architecture.

2.11 Week 11 – Integration and testing

The integration provided insights about problems that were previously concealed by the parts of the system. I have worked on the behaviour of the system, configurations, dependencies, imports, Android configuration, and modifications to the Git environment. I learned that it was impossible to understand background behavior on a physical device using the results of a test conducted in the foreground and learned that testing make it possible to identify issues that became visible during integration processes.

2.12 Week 12 – Documentation and resilience

I have made an analysis of the specifications, architecture, hazards, constraints, and assessment. As I had to explain conclusions rather than describe code, I realised that documentation was much more complicated than I believed. This has led to exposing various defects, in particular, defects dealing with the constraints of the background platform and the need for more systematic and thorough testing.

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6. Action plan

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